3D Roguelike RPG Complete System: TPS, Enemy AI, Puzzles, Camera and Props

- Documentation: Version 2 Update -

ABOUT THIS UPDATE

This document covers everything that was added in Version 2 of the package. It is written as a supplement to the main documentation, so please keep both of them together. Every feature described here is already set up in the Sample Scene, so you can open the scene and press Play to try all of it straight away.

Version 2 adds four new systems and completes one that was unfinished:

- **Inventory System** with a full UI, replacing the old one weapon plus one shield model.
- **Save and Load System** writing JSON, with a Main Menu scene and a Continue button.
- **Minimap** with coloured icons and a full screen map.
- **Speed Potion**, which is now fully working.

If you have further questions, you can contact with me via queendeveloper95@gmail.com.

INVENTORY SYSTEM

Your hero now carries a bag instead of holding a single weapon. When you walk over a weapon and press your Interact key, the weapon disappears from the world and goes into your inventory. If the matching hand is empty, it is equipped immediately. If you are already holding something, the new weapon waits in the bag until you equip it yourself.

Press **I** on the keyboard, press **Select** on your gamepad, or tap the bag button on the mobile screen to open the inventory window. The game pauses while the window is open.

- Click a weapon to equip it. Click it again to unequip it.
- Click a potion to drink it.
- Hover over any item to read what it does.
- The panel on the right always shows what you are currently wearing.

The inventory holds 20 different stacks by default. You can change this from the `Capacity` field on the Inventory System component, which sits on your Game Manager prefab.

Item Database

Every item in the game is described by an Item Definition asset, and all of them are collected in one Item Database asset. You can find them in `Assets\AdvancedRogueLikeandPuzzleSystem\Scripts\Inventory\Database`.

The inventory and the save file only remember an item Id and how many you carry. The database is what turns that Id back into a real prefab, an icon and a name. This is why saved games can restore your weapons correctly.

To add your own item to the game:

- Create the asset from the menu: **Assets > Create > Advanced Roguelike > Item Definition**.

- Give it an Id that no other item uses, a Display Name, an Icon and, for weapons, the Weapon Prefab.
- Add it to the Items list on the Item Database asset.

For a weapon, the Id must be the same as the ID on its Weapon Script component. That is how the weapon lying on the floor knows which inventory item it belongs to.

> **Item icons:** the three potions come with icons. The weapons do not, because the package has no weapon icon sprites yet. Until you add your own, weapon slots show the item name instead of a picture. To add one, import a sprite and drop it into the Icon field of the matching Item asset.

POTIONS

There are three Potions in the game. Health and Mana potions are stored in your inventory and you drink them whenever you need them. The Speed potion is new in this version.

- **Health Potion:** When you press key number 1 on keyboard or tap on potion icon on the mobile screen, it will give you 25+ Health and heal your character. You can carry four of them.
- **Mana Potion:** When you press key number 2 on keyboard or tap on potion icon on the mobile screen, it will give you 50+ Mana so you can sprint with your character by using Left Shift key. You can carry four of them.
- **Speed Potion:** Click it in your inventory and your character will move faster for 15 seconds, then return to normal speed. How much faster is taken from the `Extra_Speed` value on the Potion Script of your potion prefab, so you can tune each potion separately.

You can change the Speed potion duration from the `Speed Potion Duration` field on your Hero Controller component. The two bottle icons on your HUD always show how many Health and Mana potions you are carrying.

> **Please note:** the Shield potion and the Invisibility potion have been removed from the package in this version.

SAVE AND LOAD SYSTEM

Your game can now be saved and loaded. The save is written as a JSON file, so you can open it in any text editor while you are testing. You will find it here:

```
Windows:  C:\Users\<YourName>\AppData\LocalLow\<Company>\<Product>\arps_save.json
```

This is Unity's `Application.persistentDataPath`, so the same file works on every platform without you changing anything.

What Is Saved

- Your hero's position, facing, health, mana, coins and level.
- Your whole inventory, including both equipped items.
- All the keys you have collected.
- Which potions, keys and weapons you have already picked up, so they do not come back.
- Which chests you have already opened.
- Which doors you have left open.

When It Saves

The game saves when the player presses the Save Game button in the pause menu. It also saves automatically the first time each door is opened, which works nicely as a checkpoint. If you do not want this, untick `Auto Save On Door Opened` on the Game Session component.

Saveable Id

For the game to remember that a chest was opened or a potion was taken, every one of those objects needs its own id. This is what the Saveable Id component does. It gives the object a unique id the first time you add it, and you never have to type anything yourself.

If you **duplicate** an object in your scene, the copy starts with the same id as the original. The Game Session component checks for this when you press Play and writes an error in the Console naming both objects. Just clear the Id field on one of them and a fresh id will be created.

MAIN MENU SCENE

The package now has a Menu Scene, and it is the first scene in your Build Settings, so your game starts there. You can find it at `Assets\AdvancedRogueLikeandPuzzleSystem\Scene\MenuScene.unity`.

- **Continue:** loads your last save and puts you back exactly where you were. The button stays greyed out until a save file exists, and the date of your last save is shown at the bottom of the screen.
- **New Game:** starts from the beginning. Because this deletes your existing save, the game asks you to confirm first.
- **Quit:** closes the game.

You can set which scene New Game loads from the `Gameplay Scene Name` field on the Main Menu Controller component. Continue always uses the scene stored inside the save file, so it works even if your player saved in a different level.

The pause menu inside the game has two new buttons as well: **Save Game** and **Main Menu**.

MINIMAP

A second camera looks down at your level from above and draws it into a small map on your HUD. Anything you want to appear on that map only needs one component.

Add the **Minimap Icon** component to any prefab or scene object and choose its type. Each type has its own colour, and you can change all of them from the Minimap component:

Icon Type	Used for
Player	Your hero. The blip turns with your character so you can see which way you are facing.
Enemy	Soldiers and Archers.
Item	Potions and keys.
Door	Doors.
Chest	Chests.
Objective	Anything you want to highlight for the player.

Press **M** on the keyboard or press the **Right Stick** on your gamepad to open the full screen map. Useful fields on the Minimap component:

- **View Size:** how much of the world fits on the map. A larger value zooms further out.
- **Camera Height:** how high above your hero the map camera sits.
- **Rotate With Player:** when it is on, the map turns with your character. When it is off, north is always up.
- **Zoom In and Zoom Out:** ready to be hooked to your own buttons if you want the player to control the zoom.

CAMERA ZOOM ON UNITY 6

If you are coming from an older Unity version, you may have noticed that the mouse wheel zoom felt much slower on Unity 6. This was not your setup. Newer versions of the Input System package normalize the mouse wheel value, so the number arriving in the code is around 120 times smaller than it used to be.

The Advanced TPS Camera now handles both cases, so the zoom feels the same on every Unity version. Zoom speed is no longer buried in the formula either. You can tune it from two fields:

Field	Meaning
Zoom Speed	How far the camera moves for one mouse wheel notch.
Gamepad Zoom Speed	How fast the camera moves while the gamepad d-pad is held.

CHANGES TO EXISTING FEATURES

Your existing scenes will keep working after you update. Scene and prefab references are all kept, and inspector values you have set yourself are carried over automatically. There are only a few things worth knowing.

Renamed Scripts

Four scripts were renamed into English. You only need to change your own code if you referenced them by name.

Old Name	New Name
AsansorScript	ElevatorScript
AynaCarkScript	MirrorGearScript
DikenScript	SpikeTrapScript
TestereScript	SawTrapScript

Weapons

Picking up a weapon now puts it in your inventory instead of straight into your hand, and the copy lying on the floor is removed. This means your scene needs an Inventory System component, which is already on the Game Manager prefab. If it is missing, the Console will tell you exactly what to add.

Potion Code

`HeroController.AddPotion()` still exists and still works, but potions are now stored by the Inventory System. If your own code reads potion counts, please use `InventorySystem.Instance.CountOfPotion()` or `HeroController.HealthPotionCount` instead of the old lists.

CONCLUSION

With this update the package now has an inventory, a working save system, a main menu and a minimap on top of everything it already offered. You can build a complete Roguelike, RPG or Third Person Shooter game with it, from the title screen to the saved game, without writing any of that plumbing yourself.

I hope you like it.

Kind Regards,